Multitask Deep Learning for Brain Tumor Segmentation and Classification in MRI

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Abstract

Accurate analysis of brain tumors in magnetic resonance imaging (MRI) remains a significant challenge in clinical neuro-oncology due to the morphological complexity and heterogeneity of tumor tissue, particularly when segmentation and classification are addressed as independent tasks. To mitigate this limitation, a multitask deep learning model based on a U-Net architecture with a ResNet-34 backbone was developed and evaluated to perform simultaneous brain tumor segmentation and classification within a unified and reproducible framework. The model was trained and tested on the BRISC2025 dataset, which comprises four tumor classes and corresponding binary segmentation masks, using 5,000 images for training and 1,000 images for testing. Image preprocessing included grayscale conversion, intensity normalization, and resizing to 256 × 256 pixels. Shared feature representations were learned across tasks, with the segmentation branch producing pixel-wise probability maps and the classification branch predicting tumor categories. Supervised training employed the Adam optimizer with a combined Dice and cross-entropy loss function, complemented by early stopping, model checkpointing, and adaptive learning rate reduction to ensure stable convergence. The proposed approach achieved a classification accuracy of 97.7%, a mean Dice coefficient of 81.41%, and an intersection-over-union score of 69.55%, demonstrating accurate and consistent performance on MRI data. Overall, the multitask architecture enhances generalization and efficiency despite

its moderate complexity and provides a competitive, reproducible methodological framework for the development of clinically applicable artificial intelligence systems in brain tumor analysis.

Keywords: Brain tumor, U-Net, ResNet, Multitask deep learning, MRI segmentation

1 Introduction

Brain tumors comprise a heterogeneous group of intracranial neoplasms with diverse biological behavior, ranging from slow-growing benign lesions to highly aggressive malignant tumors. These pathologies represent a major challenge for diagnosis, treatment planning, and prognosis assessment, particularly due to their morphological variability and infiltrative nature [1, 2]. Accurate identification and characterization of brain tumors are critical for improving patient outcomes, especially in aggressive entities such as gliomas, which are associated with poor survival rates [1, 3].

Magnetic resonance imaging (MRI) is the clinical gold standard for brain tumor diagnosis and follow-up due to its excellent soft-tissue contrast and its ability to capture anatomical and functional information across different acquisition protocols [4, 5]. Despite its diagnostic value, MRI interpretation remains highly dependent on radiologist expertise and is subject to inter- and intra-observer variability, particularly in tumor boundary delineation and subtype differentiation [6–8]. These limitations motivate the development of automated methods that enhance objectivity, consistency, and reproducibility in clinical practice.

In recent years, artificial intelligence (AI) and deep learning techniques have gained prominence in medical image analysis. Deep convolutional neural networks (CNNs) have demonstrated strong performance in brain tumor classification tasks, benefiting from transfer learning and large annotated datasets [9–11]. Architectures such as VGG, ResNet, DenseNet, and EfficientNet have achieved high classification accuracy, often exceeding 95% in controlled experimental settings [12–14]. However, these models frequently rely on deep and computationally demanding architectures, which can limit their scalability and deployment in real clinical environments, particularly in low-resource settings [15, 16].

Automatic tumor segmentation has also become a central task in neuro-oncology, as precise delineation of tumor regions is essential for treatment planning and disease monitoring. Encoder–decoder architectures such as U-Net and its variants have become the dominant approach for biomedical image segmentation, achieving average Dice scores close to 0.8 in brain MRI datasets [17]. Extensions incorporating attention mechanisms, residual connections, and hybrid designs have reported improved segmentation performance [18, 19]. Nevertheless, segmentation accuracy remains sensitive to tumor heterogeneity, image quality, and acquisition variability, often requiring complex architectures or ensemble strategies that increase computational cost [20].

Several recent studies have addressed brain tumor analysis using AI-based approaches that focus either on classification or segmentation. CNN-based diagnostic support systems have demonstrated acceptable accuracy when validated by clinicians,

but they lack explicit spatial tumor localization [21, 22]. Vision Transformer-based classifiers have shown promising performance and faster inference, although their high computational requirements pose challenges for widespread clinical adoption [23, 24]. Lightweight CNN models combined with explainable artificial intelligence techniques have improved interpretability while maintaining competitive accuracy; however, their simplified architectures may limit their ability to capture complex tumor morphology [25]. Other approaches based on dense connectivity have achieved high classification accuracy through feature reuse but at the cost of increased computational demand [14]. Hybrid systems combining U-Net-based segmentation with advanced classifiers have shown strong performance and improved transparency, yet they often rely on multi-stage pipelines that reduce methodological simplicity and reproducibility [19, 25].

A common limitation across these approaches is the treatment of segmentation and classification as independent tasks, despite their strong anatomical and clinical interdependence. Tumor morphology and spatial extent provide critical cues for tumor typing, while semantic classification can guide and constrain segmentation. Addressing these tasks separately may lead to redundant feature extraction, increased model complexity, and inconsistencies between spatial and semantic predictions.

Multitask learning has emerged as an effective strategy to overcome these limitations by jointly optimizing related tasks within a unified framework. By sharing internal representations, multitask models introduce inductive bias that can improve generalization, reduce overfitting, and enhance learning efficiency [26]. In brain MRI analysis, this paradigm enables the simultaneous learning of spatial and semantic features relevant to both tumor segmentation and classification.

In this work, a multitask deep learning model is proposed for the simultaneous segmentation and classification of brain tumors in MRI images. The architecture is based on a U-Net framework with a ResNet-34 encoder backbone, allowing efficient feature sharing between a pixel-wise segmentation branch and a global classification branch [27]. The model is evaluated using the BRISC 2025 dataset, which provides aligned segmentation masks and tumor class labels across multiple anatomical planes [28–30]. By integrating both tasks into a single model, this study aims to improve efficiency, generalization, and reproducibility while maintaining competitive performance relative to state-of-the-art methods. The effectiveness of this approach is demonstrated by our experimental results (see Table 2), where the model achieves an accuracy of 97.70%, a Dice coefficient of 81.41%, and an IoU of 69.55%, effectively validating the proposed multitask framework. The main contributions of this research are as follows:

- We introduce an end-to-end multitask architecture with a shared ResNet-34 encoder and two task-specific heads, enabling simultaneous tumor segmentation and tumor-type classification from a single MRI slice.
- We propose a unified training objective that combines Dice-based segmentation loss with weighted cross-entropy for classification, and we report performance using complementary overlap metrics (Dice and IoU) together with classification accuracy.
- We provide a reproducible experimental setup on BRISC2025, including standardized preprocessing, training strategy, and evaluation metrics.

The remainder of this paper is organized as follows. Section 2 describes the dataset, preprocessing steps, and the proposed multitask model architecture, along with the training strategy and evaluation metrics. Section 3 presents the experimental results obtained for both classification and segmentation tasks on the BRISC 2025 dataset. Section 4 discusses the results in relation to existing approaches and outlines the main limitations of the study. Finally, Section 5 summarizes the main findings and highlights directions for future work.

2 Materials and methods

Data set

The BRISC2025 dataset [28] was used to develop the model. This dataset was designed for the purpose of evaluating brain tumor classification and segmentation tasks in MRI images. This set is structured so that both tasks use the same input images, which promotes the integration of results and consistency between trained models. The BRISC2025 dataset is organized into two main tasks: classification and segmentation, each with divisions for training and testing. In the classification task, images are distributed across four categories: glioma, meningioma, pituitary tumor, and non-tumor. The corresponding training/testing split and class distribution are summarized in Table 1. In the segmentation task, the dataset includes MRI slices and their aligned binary masks for tumor region delineation. All images are in .jpg format. In the segmentation task, the training and testing sets include original images (3933 for training and 860 for testing) in .jpg format, along with the corresponding binary masks (3933 and 860) in .png format, which delimit the tumor region.

Tumor category	**Training set**	**Testing set**	**Total images**	**Class Weight**
Glioma	1 147	254	1 401	1.09
Meningioma	1 329	306	1 635	0.94
No tumor	1 067	140	1 207	1.17
Pituitary	1 457	300	1 757	0.86
Total	5 000	1 000	6 000	—

Table 1 Distribution of images per tumor class, dataset split, and class weights used during training.

As shown in Table 1, a moderate imbalance is present among tumor classes. To mitigate potential bias toward more frequent categories, class weights inversely proportional to class frequency were used in the weighted cross-entropy loss.

Exploratory visualizations were generated using Matplotlib as a quality control step to confirm the match between the image and the mask and detect possible corrupt files; no inconsistencies were observed in the samples inspected.

Preprocessing and loading data

All images were converted to grayscale and resized to 256x256 pixels in order to standardize dimensions and reduce computational complexity without compromising relevant anatomical information. This size was selected as a compromise between spatial fidelity and training efficiency, following recommendations found in similar studies on brain tumors.

The input images were normalized to the range [0,1] to improve numerical stability during model training. For the segmentation masks, a small threshold value of 0.02 was applied after resizing to obtain a binary representation of the region of interest. This threshold was selected empirically after inspecting representative samples, aiming to preserve faint tumor regions while enforcing a clear separation between tumor tissue and background. As a result, binary masks were obtained, where 1 denotes the tumor region and 0 represents the background.

Subsequently, two distinct datasets were constructed from the predefined training split provided by the dataset, one dedicated to the classification task and the other to segmentation. The classification dataset associated each image with its corresponding tumor category, whereas the segmentation dataset paired the images with their respective masks delineating the tumor regions. Both datasets were loaded in mini-batches with a size of 8. During training, samples were drawn in a shuffled order (random sampling) to reduce ordering effects and improve generalization. Conversely, the validation and test sets were evaluated without shuffling to ensure deterministic and reproducible performance metrics. Data integrity was strictly maintained, ensuring no overlap between partitions and shuffling was exclusively restricted to the training process.

To verify the correct correspondence between the input images and their associated segmentation masks, representative examples from both datasets are presented in Figure 1. The images illustrate the hypointense and hyperintense signal patterns commonly associated with brain tumors in magnetic resonance imaging, while the corresponding masks highlight the delineated regions of binarized tumor tissue. In addition, the use of balanced class distributions and consistent data loading strategies helped to reduce potential bias and improve training efficiency.

Architecture of the proposed model

The proposed model integrates two complementary tasks within a single deep learning framework: brain tumor region segmentation and tumor type classification. Accordingly, we propose a multitask architecture based on a U-Net model with a ResNet-34 encoder backbone. This design enables the sharing of internal feature representations across tasks, allowing spatial features learned for segmentation to support tumor type classification within a unified architecture.

The proposed architecture consists of two branches for segmentation and classification. The segmentation branch follows a U-Net design, where the shared encoder extracts contextual features and the decoder reconstructs a pixel-wise probability map of size [B, 1, 256, 256]. In parallel, the classification branch takes the bottleneck feature map from the shared ResNet-34 encoder and applies global average pooling followed

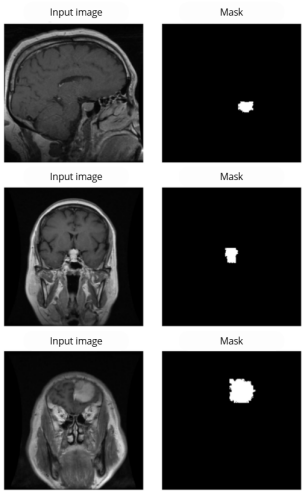

Fig. 1 Examples from the segmentation set. Left: input MRI slice (grayscale, resized to 256×256); right: ground-truth binary mask (1 tumor, 0 background).

by fully connected layers with ReLU activation and 30% dropout, ending in a 4-way softmax layer (glioma, meningioma, pituitary, and non-tumor). This branch outputs a vector of size [B, 4] with class probabilities.

During training, the model receives preprocessed MRI slices as input (grayscale, resized to 256×256, and normalized to [0,1]). The shared encoder extracts feature representations that are simultaneously fed to the U-Net decoder to predict a binary segmentation mask (tumor vs. background) and to the classification head to predict the tumor category.

In this configuration, both tasks are trained using a shared set of visual feature representations, which serves as a form of inductive bias within the model design. By relying on a common encoder, the architecture aims to reduce parameter redundancy and constrain the learning process through shared representations across tasks.

A conceptual overview of the proposed architecture is provided in Figure 2, illustrating the shared encoder and the subsequent branching into task-specific outputs for segmentation and classification.

The use of ResNet-34 as a backbone offers a balance between performance and computational efficiency, being deep enough to capture complex structural details without requiring excessive resources. Multitask training promotes consistency between spatial (segmentation) and semantic (classification) representations, which is particularly useful in medical images where tumor location is closely related to its histological type.

Model training

The model was trained using a batch size of 8 images per iteration and an initial learning rate of 1×10^{-3}. The loss function combined Dice Loss for segmentation and Cross Entropy Loss for multi-class classification, with a weighting factor of 1.0 for the classification loss.

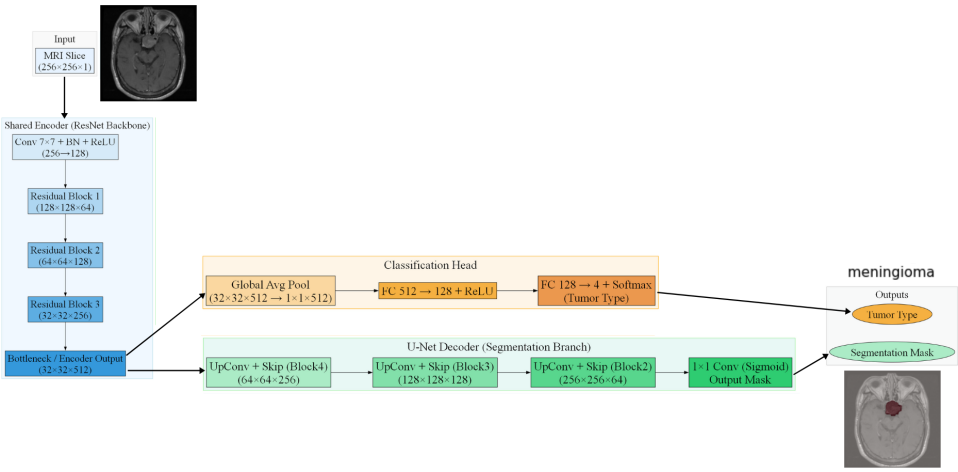

Fig. 2 Schematic overview of the proposed multitask architecture. A shared ResNet-34 encoder produces bottleneck feature maps that feed two heads: a U-Net decoder for segmentation and a classification head (GAP + FC + softmax) for tumor-type prediction.

The model was designed as a multitasking network, optimized to jointly solve the objectives of brain tumor segmentation and classification. This formulation allows the network to learn shared representations useful both for tumor localization through segmentation and for tumor type identification through classification, strengthening the overall robustness of the model.

The training process was carried out using the Adam optimizer, which adaptively adjusts the learning rate of each parameter based on the first and second moments of the gradients. Its operation is described by the following expressions. The first- and second-order moment estimates are updated according to the following equations:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{1}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \tag{2}$$

where g_t represents the gradient of the current batch, m_t and v_t are the moving estimates of the first and second order moments, and β_1, β_2 are the exponential decay factors, typically set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$.

Subsequently, the bias accumulated during the first iterations is corrected using the following expressions:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{3}$$

Finally, the model parameters are updated according to the adaptive optimization rule:

$$\theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{4}$$

where θ_t represents the model parameters at step t, α is the learning rate, and ϵ is a small value added to avoid division by zero. This procedure allows for dynamic adjustment of the update rate based on the magnitude and variability of the gradients, improving the stability and convergence of training.

To promote stable convergence during training, a ReduceLROnPlateau learning rate scheduler was employed, which monitors the validation loss and reduces the learning rate α by a factor of 0.5 when no improvement is observed over three consecutive epochs, allowing finer parameter updates as the optimization process approaches convergence.

During the training process, task-specific performance metrics were computed to monitor the optimization progress for both segmentation and classification. These metrics were aggregated on a per-epoch basis for both training and validation sets, providing a structured overview of the model behavior over time and facilitating the identification of convergence patterns as well as potential overfitting.

In conjunction with metric monitoring and adaptive learning rate adjustment, several callbacks were implemented to ensure training efficiency and reproducibility. Early stopping was applied with a maximum of 50 epochs; however, training was halted at epoch 16 when the validation loss failed to improve for seven consecutive epochs, thereby preventing unnecessary overtraining. Additionally, model checkpointing was employed to store both the best-performing model, defined by the lowest validation loss, and intermediate training states, enabling recovery and continuation of training in the event of interruptions. These mechanisms prevent both overfitting and loss of progress due to interruptions, ensuring a more stable optimization process.

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During training, the model learns iteratively by adjusting its weights to minimize total loss. In each iteration, the gradients of the loss functions are propagated backward through the network (backpropagation), and the parameters are updated by the optimizer. The optimization objective can be expressed generally as:

$$\theta^* = \arg\min_{\theta} \; \mathbb{E}_{(x,y)} \left[\mathcal{L}_{\text{total}}(f_\theta(x), y) \right] \tag{5}$$

where f_θ represents the model function parameterized by the weights θ. Over time, the model seeks to gradually reduce segmentation and classification losses, increasing its metrics (Dice and Accuracy) until it stabilizes at a point of convergence.

Training was performed on a computer with an AMD Ryzen 7 5800H processor, NVIDIA GeForce RTX 3050 Laptop GPU with 4 GB, 16 GB of RAM, a 512 GB solid-state drive, and Windows, with an approximate time of 14 hours.

Python libraries for data management, image processing, and deep learning were used for development, including NumPy, Matplotlib, scikit-learn, Pillow, and tqdm, as well as PyTorch and associated modules (torch, torchvision, torch.nn, torch.optim) and segmentation_models_pytorch for advanced segmentation architectures.

Validation

The BRISC2025 dataset provides predefined training and test partitions. To facilitate model selection and prevent overfitting, an internal validation split was created by

randomly selecting 20% of the training samples using a fixed seed. This hold-out validation strategy was used for monitoring early stopping and checkpointing. The official test set remained independent and was utilized exclusively for the final performance evaluation to ensure unbiased results.

During the validation phase, the following metrics were recorded at each epoch: segmentation loss, classification loss, Dice coefficient, accuracy, and learning rate. These variables were monitored to evaluate model performance, detect potential overfitting, and perform model selection by capturing the weights corresponding to the minimum total validation loss.

The Dice similarity coefficient, used as the main metric for segmentation, is defined as:

$$\text{Dice}(p, g) = \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i} \tag{6}$$

where p_i represents the predicted probability or binary value for pixel i and g_i is the corresponding value in the ground truth binary mask (0 or 1). Dice ranges from 0 to 1, where 1 indicates perfect overlap between prediction and ground truth and 0 indicates no overlap; therefore, higher Dice values reflect better segmentation performance.

Alternatively, the Dice coefficient can be expressed in terms of true positives (TP), false positives (FP), and false negatives (FN):

$$\text{Dice}(p, g) = \frac{2\,TP}{2\,TP + FP + FN} \tag{7}$$

To further characterize segmentation performance, the Intersection over Union (IoU), also referred to as the Jaccard Index, was computed to quantify the degree of overlap between the predicted tumor regions and the corresponding ground truth masks. IoU is defined as:

$$\text{IoU} = \frac{|\hat{y} \cap y|}{|\hat{y} \cup y|}, \tag{8}$$

where $\hat{y}$ denotes the predicted binary segmentation mask, y represents the ground truth mask, $|\cdot|$ indicates the number of pixels in the corresponding set, and the intersection and union operations quantify the overlapping and combined regions, respectively. IoU is interpreted similarly to Dice, but it typically provides a more conservative assessment of segmentation quality by penalizing mismatches more strongly.

Based on the Dice coefficient, the loss function for the segmentation task is defined as:

$$\mathcal{L}_{\text{seg}} = 1 - \text{Dice}(p, g) \tag{9}$$

For the multi-class classification task, weighted categorical cross-entropy was used, which is formulated as follows:

$$\mathcal{L}_{\text{cls}} = -\sum_{c=1}^{C} w_c\, y_c \log(\hat{y}_c) \tag{10}$$

where C is the number of classes, y_c is the true label (binary indicator for class c), $\hat{y}_c$ is the predicted probability for class c, and w_c is the weight applied to class c to correct for imbalances in the distribution of labels.

The total loss used for training and as a reference in validation is defined by combining both tasks using a weighting:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_{\text{cls}} \, \mathcal{L}_{\text{cls}} \tag{11}$$

where λ_{cls} controls the relative contribution of the classification loss with respect to the segmentation loss. The value of λ_{cls} was selected empirically based on preliminary experiments aimed at balancing the magnitude of both loss terms during training. This empirical adjustment was performed to prevent one task from dominating the optimization process, ensuring stable joint learning within the multitask framework.

Accuracy was reported as an additional classification metric, defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

All metrics cited were calculated per epoch and averaged to obtain a stable view of performance during the training and validation phases, which allowed us to identify convergence trends and detect possible signs of overfitting. The criterion for selecting the best model was the lowest value of $\mathcal{L}_{\text{total}}$ in the validation set; additionally, the metrics (Dice and Accuracy) were inspected to confirm consistency between loss and performance.

3 Results

The multitask model was evaluated using the test sets corresponding to the segmentation and classification tasks. During this phase, the model was loaded from the best checkpoint and run in evaluation mode, ensuring that no parameter updates were made. This section reports the final test performance of the proposed system for both classification (accuracy and confusion matrix) and segmentation (Dice and IoU).

Classification results

Figure 3 shows representative examples of the classification process, where the correspondence between the actual label, the predicted label, and the probability assigned by the model can be seen.

The system achieved outstanding performance in the task of classifying brain tumors, with an average accuracy of 97.70%. This result indicates that the vast majority of images were correctly classified in their respective category.

The confusion matrix, illustrated in Figure 4, shows a strong concentration of values on the main diagonal, confirming a high level of accuracy in all classes.

Segmentation results

In the segmentation task, the model achieved an average Dice coefficient of 81.41% and an IoU index of 69.55%. These values reflect a high degree of overlap between

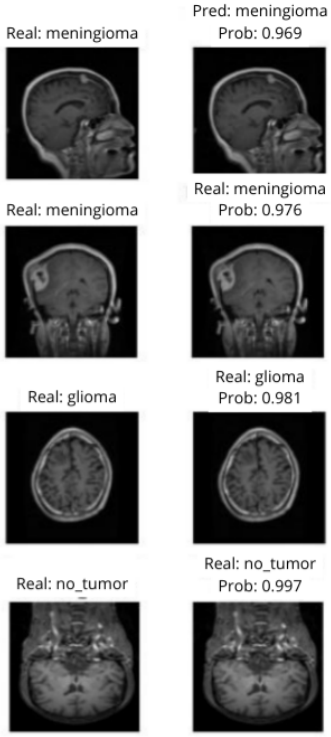

Fig. 3 Examples of brain tumor classification.

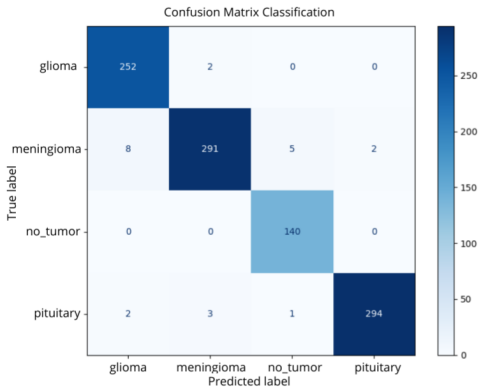

Fig. 4 Confusion matrix of the classification model.

the predicted masks and the reference masks, indicating that the model accurately delineates tumor regions in magnetic resonance images.

Figure 5 shows visual examples of the model performance on the test set. In each case, the original image, the reference mask, and the mask generated by the model are

presented. There is adequate spatial correspondence between the segmented regions and the actual tumor areas, with slight variations at the edges, especially in regions of low intensity or noise.

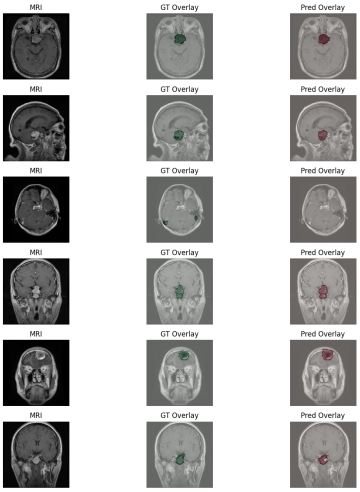

Fig. 5 Qualitative segmentation results on the test set. Left: input MRI slice. Middle: ground-truth (GT) mask overlay (reference). Right: model-predicted mask overlay (prediction).

Comprehensive performance analysis

Figure 6 shows the model ability to infer both types of results. From an input image, it can assign a label and create a binary mask delimiting the possible tumor.

The consolidated results presented in Table 2 provide a quantitative summary of the performance achieved by the proposed multitask system across both classification and segmentation tasks. At a global level, the model attains a high classification accuracy of 97.70%, indicating reliable discrimination among tumor categories as well as non-tumor cases.

Metrics	**Global**	**Meningioma**	**Glioma**	**Pituitary**	**No Tumor**
Accuracy	97.70%	99.21%	95.10%	98.15%	98.00%
Dice Coefficient	81.41%	78.80%	89.62%	73.03%	81.93%
IoU	69.55%	50.46%	83.23%	48.74%	73.31%

Table 2 Summary of global and per-class metrics of the proposed system.

From a segmentation perspective, the Dice coefficient and IoU values reflect consistent overlap between the predicted tumor regions and the corresponding ground truth masks. Notably, segmentation performance varies across tumor classes, with higher Dice and IoU scores observed for glioma cases, while comparatively lower values are

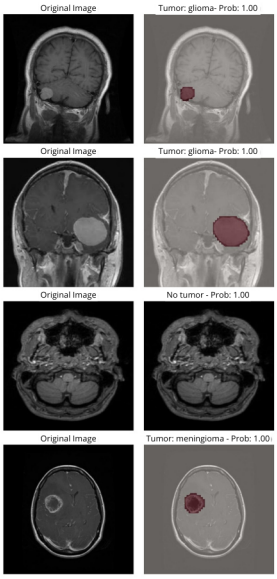

Fig. 6 Example of results from both tasks performed simultaneously.

obtained for meningioma and pituitary tumors. This variability is consistent with differences in tumor morphology, size, and boundary definition commonly observed in magnetic resonance images.

The alignment between Dice and IoU metrics across classes suggests stable segmentation behavior without pronounced inconsistencies between overlap-based indicators. Overall, the reported results indicate that the proposed multitask architecture is capable of jointly addressing classification and segmentation within a single framework, achieving balanced performance across tasks while maintaining class-specific sensitivity.

4 Discussion

The quantitative results summarized in Table 2 reveal a differentiated behavior of the proposed multitask model across tumor classes and tasks. While the classification accuracy remains consistently high for all categories, segmentation performance exhibits noticeable variability, particularly when comparing glioma cases with meningioma and pituitary tumors. This discrepancy can be attributed to differences in tumor morphology and boundary definition, as gliomas often present more heterogeneous but spatially extended regions, whereas meningiomas and pituitary tumors tend to be smaller and exhibit less well-defined margins in magnetic resonance images. These characteristics directly affect overlap-based metrics such as Dice and IoU, highlighting the sensitivity of segmentation performance to anatomical and structural variability.

In the broader context of recent deep learning studies on brain tumor analysis, Table 3 summarizes representative approaches and highlights that many methods

report results under different datasets and experimental settings, typically optimizing architectures for a single objective. Segmentation-oriented methods frequently incorporate fully convolutional designs, multiscale feature extraction, attention mechanisms, or saliency-driven components to better capture tumor boundaries, particularly in heterogeneous cases. These strategies generally improve overlap metrics, especially for tumors with diffuse or irregular morphology, but at the cost of increased architectural complexity and task-specific design. In contrast, the proposed multitask formulation integrates tumor localization and semantic discrimination within a unified architecture, allowing shared feature representations to support both tasks and providing segmentation and classification outputs simultaneously.

Ref	**Architecture / Approach**	**Dataset**	**Task**	**Accuracy**	**Dice**	**IoU**
[31]	ResNet18 / MobileViTV2 / EfficientNet-B0 (classification only)	BRISC	Cls	96.94%	–	–
[32]†	Deep learning-driven non-contrast MRI (CNN + FCN)	Other	Seg/Cls	98.3%	93.7%	–
[33]†	Automated multi-class MRI tumor model (MS-DAM)	Other	Seg/Cls	96.5%	91.6%	–
[34]†	Multiscale CNN (multiscale pathways)	Other	Seg/Cls	97.3%	82.8%	–
This work	U-Net + ResNet-34 (multitask)	BRISC	Seg/Cls	97.70%	81.41%	69.55%

Table 3 Comparison of representative approaches for brain tumor MRI analysis. Dashes indicate metrics not reported in the corresponding studies. †Results are reported under different datasets and experimental protocols and are therefore not directly comparable to BRISC.

An important aspect of the present study, as reflected in Table 3, is the explicit evaluation of segmentation performance using the Intersection over Union (IoU) metric in addition to Dice. While Dice score is widely adopted in brain tumor segmentation studies and provides a robust measure of overlap, it tends to be less sensitive to boundary mismatches, particularly in cases involving small or irregular tumor regions. In contrast, IoU penalizes false positives and false negatives more strictly, offering a complementary and often more conservative assessment of segmentation quality. However, although several recent approaches report high Dice scores, the majority do not include IoU in their evaluation protocol, which limits the ability to fully assess segmentation robustness and spatial precision, especially in cases where tumor boundaries are diffuse or when partial over-segmentation occurs.

5 Conclusions

This study presented a multitask deep learning model based on a U-Net architecture with a ResNet-34 encoder for the joint segmentation and classification of brain tumors in magnetic resonance imaging. The proposed approach achieved competitive performance, with a Dice coefficient of 81.41%, an Intersection over Union of 69.55%, and an overall classification accuracy of 97.70%, demonstrating its capacity to integrate semantic discrimination and spatial delineation within a unified framework.

Segmentation performance was assessed using both Dice and Intersection over Union metrics in order to provide a complementary and more complete characterization of overlap quality. While Dice remains the most commonly reported measure in related studies, the inclusion of IoU offers an additional perspective on boundary agreement and false positive penalization, contributing to a more transparent interpretation of segmentation results, particularly in cases involving complex tumor morphology.

Several limitations of the present work should be acknowledged. The proposed model operates on two-dimensional image slices, which may restrict the exploitation of volumetric contextual information available in three-dimensional MRI data. Moreover, the evaluation was conducted on a single dataset with a fixed experimental split, limiting conclusions regarding generalization across different acquisition settings. In addition, the weighting between segmentation and classification losses was determined empirically and may not represent an optimal balance for all tumor classes.

Future research will focus on extending the proposed framework toward three-dimensional and multimodal MRI analysis, as well as exploring adaptive loss weighting strategies and uncertainty-aware learning to improve robustness and generalization. These directions aim to further assess the applicability of multitask learning approaches in clinically realistic scenarios.

Declarations

Acknowledgments. This research was supported by the Universidad de San Buenaventura, Colombia, Cali under the project Industria 4.0: Integración de tecnologías impulsoras de la industria 4.0.

Ethical approval. Not applicable.

Authorship contribution statement. **Authorship contribution statement.** Juan D. Yépez Vélez and Jesús D. Gelves Cajiao (Conceptualization, Methodology, Software development, Validation, Formal analysis, Data curation, Writing—original draft, Visualization), and Carlos M. Paredes and José F. Valen- (Research, Writing—review and editing, Acquisition). , and Carlos M. Paredes and José F. Valencia (Conceptualization, Formal Analysis, Research, Writing—review and editing, Supervision, Project management).

Financing. The authors declare that no funding was received for the conduct of this study.

Conflict of Interest. The authors declare that there are no conflicts of interest related to the publication of this work.

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